

TASK 5 - NETWORK COMMANDS

1. Ping

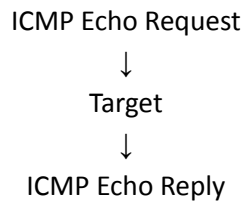
Purpose

Tests IP network reachability and measures round-trip time to a destination.

Protocol

ICMP - Internet Control Message Protocol

Basic Process



Example : ping 192.168.1.1

Common Output

- Reply from -> destination responded
- Time -> round-trip time
- TTL -> Time To Live value

Important Point

A failed ping does not necessarily mean the host is offline because ICMP may be blocked or filtered.

Cybersecurity Relevance

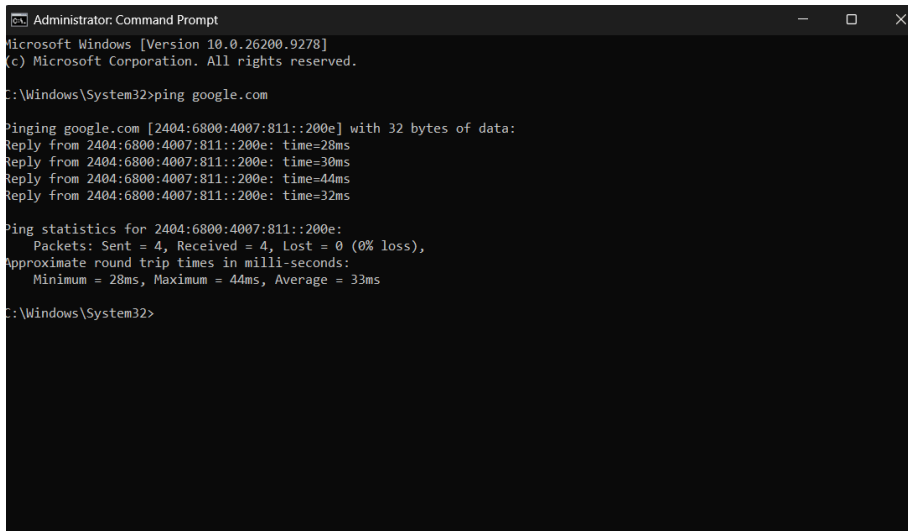
- Network troubleshooting
- Basic host reachability testing
- ICMP traffic can appear during network discovery/monitoring

PRACTICAL TASK - ping [google.com](https://www.google.com)

Command: ping [google.com](https://www.google.com)

Purpose : Test IP network reachability and measure round-trip time.

Observed : DNS resolved [google.com](https://www.google.com) to an IPv6 address, and ICMP Echo Requests received successful Echo Replies with 0% packet loss.

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window shows the output of the command "ping google.com". The output indicates that the DNS resolved google.com to an IPv6 address [2404:6800:4007:811::200e]. Four ICMP Echo Requests were sent, all received successful replies with 0% packet loss. The round-trip times in milliseconds are: 28ms, 30ms, 44ms, and 32ms. The statistics show 4 packets sent, 4 received, and 0 lost (0% loss). The approximate round-trip times are: Minimum = 28ms, Maximum = 44ms, Average = 33ms.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ping google.com

Pinging google.com [2404:6800:4007:811::200e] with 32 bytes of data:
Reply from 2404:6800:4007:811::200e: time=28ms
Reply from 2404:6800:4007:811::200e: time=30ms
Reply from 2404:6800:4007:811::200e: time=44ms
Reply from 2404:6800:4007:811::200e: time=32ms

Ping statistics for 2404:6800:4007:811::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 44ms, Average = 33ms

C:\Windows\System32>
```

2. ipconfig

Purpose

Displays IP network configuration on Windows.

Basic Command

ipconfig

Shows commonly

- IPv4 addresses.
- Subnet mask
- Default gateway.

Detailed Command

ipconfig /all

Can additionally show:

- MAC/Physical Address
- DHCP status
- DNS servers
- More adapter configuration

Key Points

- `ipconfig` -> view network configuration
- `ipconfig /all` -> detailed configuration
- Useful for troubleshooting IP connectivity.
- Connects IP, subnet masks, DHCP, and DNS concepts.

Quick Revision

`ipconfig` displays the device's IP configuration and is commonly used for basic network troubleshooting.

PRACTICAL TASK - `ipconfig`

Command : `ipconfig`, `ipconfig /all`

IPv4 : 192.168.45.1

Subnet Mask : 255.255.255.0 (/24)

Default Gateway : Not shown for this adapter

DHCP : Enabled

DNS : 192.168.1.1 (also IPv6 link-local DNS entries present)

```
Wireless LAN adapter Wi-Fi 2:
Connection-specific DNS Suffix . . . : 
Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz
Physical Address. . . . . : 38-7A-0E-D6-D2-D3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
IPv6 Address. . . . . : 2401:4900:88dc:8c8a:a68f:bc8e:6b5c:2b8f(Preferred)
Temporary IPv6 Address. . . . . : 2401:4900:88dc:8c8a:80eb:413d:3a48:4304(Preferred)
Link-local IPv6 Address . . . . . : fe80::9365:7e0f:747c:be9b%8(Preferred)
IPv4 Address. . . . . : 192.168.1.5(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 04 September 2026 21:52:44
Lease Expires . . . . . : 06 September 2026 18:06:00
Default Gateway . . . . . : fe80::a291:caff:feac:dc90%8
                             192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 154696206
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-B3-1F-B3-5C-60-BA-BE-97-44
DNS Servers . . . . . : fe80::a291:caff:feac:dc90%8
                             192.168.1.1
                             fe80::a291:caff:feac:dc90%8
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 38-7A-0E-D6-D2-D7
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . . : 
Description . . . . . : Realtek Gaming GbE Family Controller
Physical Address. . . . . : 5C-60-BA-BE-97-44
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
```

3. ifconfig

Full form : Interface Configuration

Purpose : Used mainly on Linux/Unix systems to display and configure network interfaces.

Can show

- Interface name
- IPv4/IPv6 address
- Subnet mask
- MAC address
- Interface status
- Statistics

Example

ifconfig

Windows comparison

Windows -> ipconfig

Linux/Unix -> ifconfig

Key Point

ifconfig is a Linux/Unix network-interface command, while ipconfig is the corresponding common Windows command.

PRACTICAL TASK - ifconfig

Command : ifconfig

Interface : eth0

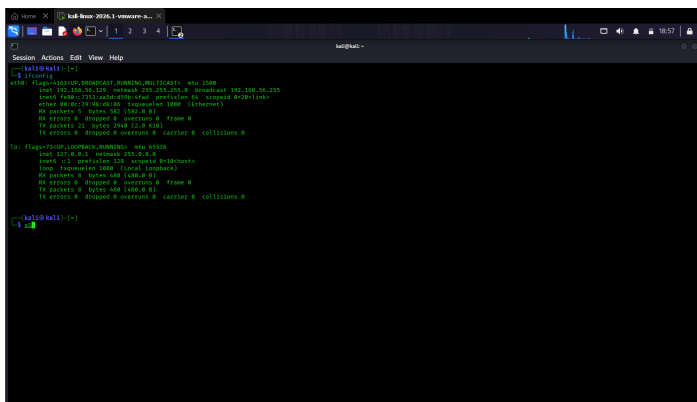
IPv4 Address : 192.168.56.129

Subnet Mask : 255.255.255.0

MAC Address : 00:0c:29:98:ad:86

Status : UP, RUNNING

Loopback : 127.0.0.1 (lo)



```
kali@kali: ~$ ifconfig
eth0: flags=4096<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.129 netmask 255.255.255.0  hwaddr 00:0c:29:98:ad:86
    inet6 fe80::c29:98ad:f43d:129a%eth0 prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:98:ad:86  txqueuelen 1000  (Ethernet)
    RX packets 0  bytes 0 (0.0 B)
    TX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128  scopeid 0x100<host>
    ether 00:00:00:00:00:00  txqueuelen 1000  (Local Loopback)
    RX packets 0  bytes 0 (0.0 B)
    TX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

4. tracert

Purpose

tracert is a Windows command used to trace the path/hops packet take toward a destination.

Command

tracert [google.com](https://www.google.com)

Output

- Each row represents a network hop.
- Shows response times for each hop.
- Shows the path toward the destination.

ping vs tracert

ping

- Tests reachability
- Measures RTT

tracert

- Shows network path/hops
- Shows response times per hop

Important Point

*** does not necessarily mean a router or destination is down; responses may be filtered or suppressed.

Key Point

tracert helps identify the sequence of network hops between your computer and a destination.

PRACTICAL TASK - tracert

Command : tracert [google.com](https://www.google.com)

Destination : [google.com](https://www.google.com)

Protocol observed : IPv6

Hops : 12

First hop : 2401:4900:88dc:8c8a:a291:caff:feac:dc90

Timeouts : Hop 2 showed * * * Request timed out

Results: Trace completed successfully at hop 12.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9278]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>tracert google.com

Tracing route to google.com [2404:6800:4007:821::200e]
over a maximum of 30 hops:
  0  2 ms  2 ms  2 ms  2401:4900:88dc:8c8a:a291:caff:feac:dc90
  1  *      *      *      Request timed out.
  2  26 ms  28 ms  27 ms  fc00:11::cb
  3  46 ms  27 ms  27 ms  2404:a800:3a00:1::61
  4  40 ms  27 ms  27 ms  2404:a800::92
  5  *      *      *      Request timed out.
  6  30 ms  27 ms  29 ms  2404:6800:8202:40::1
  7  33 ms  28 ms  32 ms  2001:4860:0:1::136a
  8  61 ms  35 ms  24 ms  2001:4860:0:1::40e8
  9  39 ms  27 ms  31 ms  2001:4860:0:1::1841
 10  48 ms  28 ms  26 ms  2001:4860:0:1::48db
 11  53 ms  26 ms  24 ms  maa05s26-in-x0e.le100.net [2404:6800:4007:821::200e]
 12

Trace complete.

C:\Windows\System32>
```

5. traceroute

Purpose : traceroute is a Linux/Unix command used to trace the network hops between a source and destination.

Command : traceroute [google.com](https://www.google.com)

Common OS comparison

Windows - tracert

Linux / Unix - traceroute

Output

- Each numbered line represents a hop.
- Shows response times for responding hops.
- *** may indicate no response/filtering/rate limiting.

Key Point

traceroute traces the path from the local system to a destination by identifying intermediate network hops.

PRACTICAL TASK - traceroute

Command : traceroute [google.com](https://www.google.com)

Operating System : Kali Linux

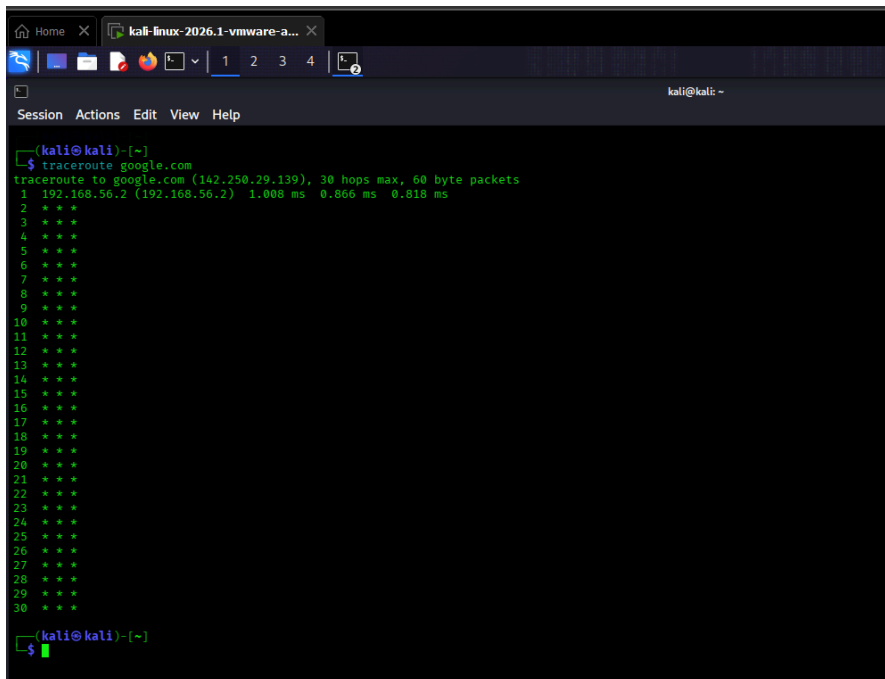
Destination : [google.com](https://www.google.com) (142.250.29.139)

Maximum hops : 30

Hop 1 : 192.168.56.2 responded successfully

Hops 2-30 : * * * - no response shown

Result : Destination was not reached/identified within 30 hops.



```
(kali@kali)-[~]
$ traceroute google.com
traceroute to google.com (142.250.29.139), 30 hops max, 60 byte packets
 1  192.168.56.2 (192.168.56.2)  1.008 ms  0.866 ms  0.818 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```